Math515 – Financial Mathematics II

Spring 2014

Homework 1

1. A and B are two independent events with $P(A) = 0.5$, $P(B) = 0.7$. Find $P(A \cup B)$, $P(B|A)$, and $P(B \backslash A)$.

2. In a school, 60% of students have access to the internet at home. A group of 8 studenst are chosen at random. Find the probabilities that

   (a) exactly 5 students have access to the internet.

   (b) at least 6 students have access to the internet.

3. Two cards are dealt from a deck. Find the conditional probability that one of the cards is the king of hearts given that at least one of the cards is a king.

4. Let X be a random variable with normal distribution $N(\mu, \sigma^2)$, prove that $E[X] = \mu$, $\mathrm{Var}(X) = \sigma^2$, and the second moment $E[X^2] = \mu^2 + \sigma^2$.

5. For a nonnegative random variable X that is with $X(\omega) \geq 0$ and a constant $a > 0$, prove

   (a) *the Markov inequality*

   $$P(\{\omega : X(\omega) \geq a\}) \leq \frac{1}{a} E[X].$$

   (b) *the Chebyshev inequality*

   $$P(\{\omega : |X(\omega)|^2 \geq a\}) \leq \frac{1}{a} E[X^2]$$

6. Suppose that a certain stock is equally likely to go up 1 unit or keep unchanged or go down 1 unit, and that the outcomes of different periods are independent. Let X be the amount of the stock goes up (either 1 or 0 or -1) in the first period, and let Y be the cumulative amount it goes up in the first three periods. Find $\mathrm{Corr}(X, Y)$.

7. For any pair of random variable X and Y, show that

   $$|\mathrm{Corr}(X, Y)| \leq 1.$$

8. Assume that the random vector (X_1, X_2) has density f_{X_1,X_2} with marginal densities f_{X_1} and f_{X_2}, prove that $E[X_1 X_2] = E[X_1]E[X_2]$ if X_1 and X_2 are independent .

9. Let $(X_t, t \in [0, +\infty))$ be the homogeneous Poisson process, compute its covariance function $c_X(t, s)$ and variance function $\sigma_X^2(t)$.